

Research Notes

1 Introduction

Document will outline my approach to learning hydraulics and how to create a hydraulic launcher. Hydraulic launchers are interesting because they are used often in rollercoaster launch systems, see figure 1, military drones, and aircrafts. Often, large drones are catapulted with hydraulic launchers because of their ability to deliver high force, and precise speed in a compact area.



Figure 1: *Wicked* rollercoaster from theme park in Utah [1].

I'll detail in this report the exact mechanics of hydraulics and how one can fabricate one from scratch.

2 Hydraulics

Hydraulics move heavy machinery, like construction equipment, around using trapped liquid. That might be ambiguous, but that trapped liquid compresses and exerts massive forces due to pressure. A good example is taking an empty syringe, placing one's finger on the end, and then pushing the plunger until it is physically unable to. When applied force on the plunger is released, the plunger springs back to its original position due to pressure. However, if that syringe is filled with water, the whole scenario becomes widely different, leading us into *Pascal's Law*, the direct mathematical foundation for hydraulics. But, first. let's analyze the physical properties of pressure and its relationship to energy.

2.1 Pressure

Pressure is defined as force per unit area, or:

$$P = \frac{F}{A} \quad (1)$$

and has the SI units Pascal, see figure 2, which are Newtons per square meter. It is often more convenient to use pressure rather than force to describe fluid behavior as they are more complicated systems.

For any object sitting on Earth, it experiences the force of gravity, and thus, the normal force exerted by the surface that it is resting on. That force of gravity is an object's weight, and the normal force is the feeling of having weight. However, for non-point masses, different orientations may yield different areas in contact with the surface, which yields different exerted pressures. Consider a 20 kg box with length 5 metres, width 10 meters, and height 2 meters. If you lay the box on its large side, the area in contact is 50 m^2 , and has a pressure computed out to:

$$P = \frac{200 \text{ N}}{50 \text{ m}^2} = 4 \text{ Pa}$$

But, if the box is laid on a more narrow side, the area in contact is 20 m^2 and the pressure then becomes 10 Pa . Intuitively, it makes sense why the pressure would be more. Standing on our feet, the pressure is much more manual, but if you try to balance on one finger, even just for a second, the pain on that hand becomes unbearable. When the area of contact decreases, the pressure increases. This explains why sharper objects cut easier with less amounts of force. When the tip of a blade is sharpened so that the area of contact is as small as possible, it cuts through materials like butter.

For non-solid objects, pressure is considered differently. For liquids, the medium is treated as a continuous distribution of matter, whereas a gas' pressure is considered to be the average pressure from molecular collisions with the wall. Think *Ideal Gas Law*: $PV = nRT$. When the number of moles increases, there are more gas molecules hitting the wall, increasing the pressure

Due to the nature of equation (1), adding another distance multiplier can change the nature of the equation. Let's consider the box example again. Remember that work is defined as the force applied parallel to the displacement of an object, or:

$$W = F \cdot d \quad (2)$$

and that if adding an additional displacement modifier to area makes the unit m^3 , indicative of volume, where:

$$V = A \cdot d$$

then, one can substitute both of these equations into the numerator and denominator respectively for equation (1), so that:

$$P = \frac{F \cdot d}{A \cdot d} = \frac{W}{V} \quad (3)$$

where the displacement vector cancels. Then, by the definition of the third term in equation (3), pressure can be defined as a measure of the energy per unit volume of an object. Since energy is *stored* per unit volume when pressure is applied, that alludes to the fact that there is a kinetic and potential energy associated with fluids.

If pressure is able to be defined by the energy per unit volume of a liquid, then pressure should be able to be defined as kinetic energy per unit volume of a liquid, as well as potential energy per unit volume of a liquid. If a liquid is moving, then the standard kinetic energy equation for moving objects, being:

$$KE = \frac{1}{2}mv^2 \quad (4)$$

can be substituted into the W portion of equation (3), so that:

$$P = \frac{\frac{1}{2}mv^2}{V}$$

Remember that density is described by how much mass fits into some specific volume, or mathematically:

$$\rho = \frac{m}{V} \quad (5)$$

which can then be substituted in the prior equation where:

$$P = \frac{1}{2}\rho v^2 \quad (6)$$

Furthermore, one can define pressure in terms of potential energy per unit volume as well, which can be used by deriving the potential energy for a mass suspended midair. If work equals the force acting parallel to the displacement to an object, as seen in equation (2), and if intuitively, the object does fall some amount of height h , then the potential energy stored in that object is:

$$U = mg \cdot h \quad (7)$$

which can then be substituted into equation (3). The reason this is the case is because, technically, water molecules are subject to gravity and will tend to *fall*, thus, storing energy. With further consideration to the density equation (5), pressure is defined as:

$$P = \rho gh \quad (8)$$

This equation is also known as the hydrostatic pressure of a liquid, which is the pressure exerted by a fluid at rest or in equilibrium, caused by the force of gravity. It is the weight of a liquid pushing down on itself and all other walls that surround it, and explain why it is unsafe to go deep into the ocean because hydrostatic pressure becomes so high. It also explains why one's ears will pop if they dive into water. When going deeper into water, the hydrostatic pressure increases overhead, causing a subsequent equalizing reaction of the ears to balance pressure.

2.2 Pascal's Law



Figure 2: French mathematician, philosopher, physicist, and prose writer; discovered Pascal's Law [2].

Considering the syringe scenario again, when the syringe is instead filled with a liquid such as water, the fluid is not as compressible in comparison to air, and begins to exert its pressure forces on the exterior of the syringe. Pascal's law restates this phenomena, claiming that any pressure applied to a confined liquid has its value project equally and undiminished (does not weaken) in all directions to every part of the whole fluid and the walls of its container.



Figure 3: Pressure is applied due to the stopper pushing down on the bottle [3].

Imagine a champagne bottle, like one seen in figure 3, has a stopper applying a constant force to keep the champagne contained. Let's say that the force the stopper is exerting on the tip of the bottle is equal to 10 N , and the area is 5 cm^2 . Therefore, with help from equation (1), the pressure at the tip is equal to $20,000\text{ Pa}$ or 20 kPa . With attention to Pascal's Principle, the pressure should then be exerted to every part of the liquid as well as the system that encloses it. Then, the sides of the bottle also experience a pressure equal to 20 kPa . What, though, is the force exerted on the sides of the bottle.

Let's consider the bottom of the bottle to have a pressure of 20 kPa with a surface area equal to 50 cm^2 . Rearranging equation (1) for force gives us:

$$F = PA \quad (9)$$

where the force at the bottom of the bottle is equal to 100 N . Wait, the stopper of the bottle was exerting a force of 10 N , how is it possible that there is more force at the bottom of the bottle than at the top. Remember that pressure goes undiminished, not force, meaning that there can be different forces at different parts of the bottle with the same pressure. Pascal actualized this through an experiment with a long narrow tube attached to a barrel. When the tube's height increased, the hydrostatic pressure increased as well, which was consistent with equation (8). However, as height increased, the barrel became more subject to bursting, until it did. Hence, Pascal realized that pressure was increasing throughout the entire system, but the force that was applied to the walls of the barrel changed.

Knowing that force becomes multiplied when pressure is the same for a container, but different for specific areas of the container, is the driving *force* for the concept of hydraulics. Moreover, it should be noted that at the bottom of any fluid container, there is some hydrostatic pressure overhead due to all the water molecules affected by gravitational potential energy. The pressure at the bottom of the container should then become:

$$P_{bottom} = P_{top} + \rho gh \quad (10)$$

2.3 Hydraulic Press

Now, we'll analyze a common hydraulic press scenario to illustrate the mathematics behind them. Consider an apparatus such as one seen in figure 4, where there is a hydraulic press attempting to move a car. m_{car} , d_1 , and d_2 are given, and we are asked to evaluate what force F_{piston} is required to move the car a distance d_2 .

Due to the conservation of energy principle, we can assert that the energy input of the system is equal to the energy output of the system, with the neglecting of non-conservative forces, so that:

$$W_{in} = W_{out} \quad (11)$$

And, with the equation (2), this equation expands out to:

$$F_{piston} \cdot d_1 = m_{car}g \cdot d_2 \quad (12)$$

where the right-hand side is gained by the acknowledgement that the weight of the car is being moved by a distance d_2 . The dot products of the forces and distances can be replaced by standard multiplication since the force is acting directly parallel to the displacement, and moreover, the equation (12) can be manipulated to solve for F_{d_2} where:

$$F_{piston} = m_{car}g \frac{d_2}{d_1} \quad (13)$$



Figure 4: Hydraulic press lifting a car (excuse the crude drawing).

This introduces a very important fact about hydraulics, in that, if you want F_{piston} to be small (in order to export a large amount of force), you must make it so that d_1 is a large value. If we introduce some numbers and say that the car is a *2025 Toyota Camry* with a mass approximately 1600 kg , then, in order to move the car up 5 m , with d_1 of 30 m , the force applied to the first piston should equal 2613 N . Although this force is noticeably smaller than the weight of the car, it is still around the weight of a grizzly bear. However, if we make it so d_1 is instead 100 m , then the force applied to the first piston becomes equal to 784 N .

This is still not what we want! If we want hydraulics to be powerful, we should derive another way to decrease the force applied to make the *2025 Toyota Camry* to move upwards by 5 m . With that, we'll claim that r_1 and r_2 radii of the pistons respectively are given, and that the pistons have the surface area conforming to the area of a circle. Due to the equation (1), there is a pressure exerted undiminished and equally to every part of the system by the car, where:

$$P_{car} = \frac{m_{car}g}{\pi r_2^2} \quad (14)$$

Then, there is some force being exerted on the surface area of piston #1 where:

$$F_{piston} = P_{car}\pi r_1^2$$

which after have its pressure term expanded due to equation (14), the force applied to the piston becomes:

$$F_{piston} = m_{car}g \frac{r_1^2}{r_2^2} \quad (15)$$

Let's lock in the F_{piston} in equation (15) to be 100 N , equivalent to lifting a large watermelon. Let's then revert the initial distances d_1 and d_2 to 30 m and 5 m respectively. Then, it seems intuitive to change r_1 and r_2 to get the intended 100 N I desire.

For incompressible fluid, any push in any direction must be followed by an equal and opposite push in a different direction. In other words, because there is no compressing, the mass of the fluid stays the same, and because the amount of fluid doesn't change, the volume of the fluid should not either. This concept is known as the conservation of volume, which holds that:

$$V_{in} = V_{out} \quad (16)$$

So, in this hydraulic press scenario, the volume changes that occur when a piston is pushing down are equal to the cross-sectional area times the displacement of the piston. If the first piston is pushed down, then the second piston is pushed up, which yields a corresponding volume change as well. Thus, using equation (16), we can see that both volume changes must equal each other to satisfy the conservation of volume, where:

$$\pi r_1^2 d_1 = \pi r_2^2 d_2 \quad (17)$$

which yields a non-linear equation where r_1 and r_2 must satisfy:

$$\frac{r_2^2}{r_1^2} = \frac{d_1}{d_2} \quad (18)$$

Now, that means there is some constraint to the situation, and that, we cannot get exactly 100 N pushing down on the piston. We will always get a F_{piston} that is consistent with a computation from the equation (13), and that the smallest possible force I can get is constrained by the distances the pistons travel by, which is actually a good conclusion to changing design constraints.

2.4 Hydraulic Brakes

Hydraulic brake calipers, like one seen in figure 5, use pressurized fluid to push internal pistons and force brake pads to clamp onto the spinning rotor driving the wheel. When the brake pads clamp on to the rotor, a friction force is applied directly opposite to the linear motion of the motor, and applies a torque corresponding to the lever arm it is applied on.

Whenever one presses a brake pedal in the car, there is a mechanical signal for an engine vacuum or auxiliary pump to multiply the force applied to a piston in a master cylinder filled with hydraulic fluid. The piston in the hydraulic fluid creates pressure equally and undiminished in all directions that travels in tubes that route to the brake caliper. Inside the caliper are small mechanical pistons attached to brake pads that then push inwards and apply force to the rotor. Let's reverse engineer the amount of force that the piston in the master cylinder is applying to achieve stoppage for a car traversing 60 mph . Considering the brilliant engineering of *Brembo* brakes, 60 to 0 stoppage can a very short amount of time. We can then derive the average force applied during that amount of time by utilizing the impulse equation (21). Impulse as the total

change in momentum caused by a force applied over some specific amount of time. This comes from the characterization that force is defined as the change in momentum over the change in time, or:

$$F = \frac{dp}{dt} \quad (19)$$

which is intuitive as force causes acceleration, a change in velocity, with respect to the inertia of that object, or mass. And, since momentum is an object's velocity times its inertia, then a force will cause a change in that momentum. So, considering equation (19), we can solve for dp and get:

$$dp = F dt \quad (20)$$



Figure 5: *Brembo* is a world leader and pioneer in disc brake technology for the automotive industry [4].

From this equation, we can claim that F can be simplified by redefining it as the average force applied during dt , or, F_{avg} . Furthermore, dp and dt can be changed to more macroscopic variants where:

$$\Delta p = F_{avg} \Delta t$$

where if p is expanded out to its definition, the above equation becomes:

$$m(v_{final} - v_{initial}) = F_{avg} \Delta t \quad (21)$$

assuming that the mass of the object does not change. Then, rearranging for F_{avg} yields:

$$F_{avg} = \frac{m(v_{final} - v_{initial})}{\Delta t} \quad (22)$$

Therefore, this becomes the amount of force needed to completely halt the car with some initial velocity. However, there is a caveat to this equation. F_{avg} is the force acting directly against linear motion, but is not the force that is acting upon the entire car. If this were true, then there would be one mega brake for the entire car, but each wheel usually has its own brake.

Thus, we can use F_{avg} to develop a smaller braking force, specifically for the front wheels, by the equation:

$$F_{wheel} = \frac{\phi F_{avg}}{2} \quad (23)$$

where ϕ is the braking share between the front and back wheels which will be denoted by ϕ_{front} now, and the entire equation is divided by two since both front wheels endure the force and exert a braking force. Furthermore, we must introduce the torque τ which more accurately describes the situation at hand. When cars move, they are not only advancing further, but are having its wheels rotate to propel itself forward. Therefore, there must be some additional energy at hand to halt the car as well as to stop rotation of the wheels. Torque is the measure of an object's angular acceleration coupled with its inertia. However, the inertia for objects with rotational motion is not mass, but rather, the moment of inertia.

Moment of inertia is the measure of an object's resistance to changes in rotational speed. We can find the simple definition of moment of inertia by manipulating the kinetic energy equation (4). Rotational velocity can be related to linear velocity by claiming that linear dimensions are related to angular dimensions through the radius and angle. For example, rotational displacement θ times a radius r yields an arc length l . That arc length can be equated to linear displacement d if it is stretched out like a tangent line. An example is that when a wheel of a tire rotates, the corresponding distance traveled is equal to the arc length corresponding to the radius of the tire. So, this can be extrapolated to other forms of linear motion, including velocity, making the kinetic energy equation:

$$KE = \frac{1}{2}mr^2\omega^2 \quad (24)$$

We can then take the term mr^2 and define it as I . For linear kinetic energy, the inertia term is m , but for angular energy it is I . Then we can think of these two quantities as the inertia term corresponding to the specific type of motion. However, I is not defined as mr^2 , but rather, it is defined as the inertia at every single radius pertaining to the object that is rotating. Therefore, the moment of inertia should be equal to the sum of inertias throughout the radius, or:

$$I = \int r^2 dm \quad (25)$$

Let's model the internal motor as a simple disk with mass m . The disk has an areal mass density of σ that is defined as:

$$\sigma = \frac{m}{A} \quad (26)$$

Then, we can rearrange the prior equation for m , and take the basic derivative of all components, such that:

$$dm = d\sigma dA \quad (27)$$

and substitute into equation (25) to get:

$$I = \int r^2 d\sigma dA \quad (28)$$

Then, we must evaluate dA in terms of r so we can integrate the r^2 term. The area of a circle is defined as $A = \pi r^2$, where the derivative becomes:

$$dA = 2\pi r dr \quad (29)$$

Let's also define the lower limit and the upper limit of the integral in equation (28) as 0 and r respectively, since we are evaluating the moment of inertia at every point of the circle from the center to its edge. This allows us to take the $d\sigma$ term out of the integral because the mass density does not vary by r , and the integral is from the center to the edge, containing the whole of σ . Finally, if we substitute the equation (29) into the updated integral we get:

$$I = 2\sigma\pi \int_0^r r^3 dr \quad (30)$$

Then, taking the integral yields:

$$I = \frac{1}{2}\sigma\pi r^4$$

Our work is still not done, though. Since areal mass density has a definition, given by equation (26), we can further simplify the above equation and get:

$$I = \frac{1}{2}mr^2 \quad (31)$$

Thus, we can claim that Newton's 2nd Law applies to situations pertaining to rotational momentum, where:

$$I\alpha = \Sigma\tau = T - F_{wheel}R \quad (32)$$

where R is the radius of the tire. The reason for including the second term on the right-hand side is that the road exerts an additional torque associated with moving forward. T is the term that describes the braking torque exerted by the brake caliper. Furthermore, one can solve for T using equation (32) and yield:

$$T = F_{wheel}R + I\alpha \quad (33)$$

So, the braking force must satisfy the above equation in order to effectively stop the car. Let's denote the net torque of (32) as T for simplicity sake, and claim that the clamping force of the caliper is N , where it is pinching the internal rotor at a radius r_{pad} . Then, the equation relating the two quantities is as follows:

$$T = 2\mu_k N r_{pad} \quad (34)$$

where the force is doubled since clamping yields two forces acting on the same object, and μ_k is the friction coefficient associated with the rotor and the brake pads. Then, N can be isolated, T can be expanded via equation (32), and α can be written out as a rotational kinematic equation solving for angular acceleration to get:

$$N = \frac{F_{wheel}R + I\left(\frac{\omega_{final} - \omega_{initial}}{\Delta t}\right)}{2\mu_k r_{pad}}$$

and then, when F_{wheel} is expanded, as well as the moment of inertia, the equation yields:

$$N = \frac{\phi_{front} m_{car} (v_{final} - v_{initial}) R + m_{rotor} r_{rotor}^2 (\omega_{final} - \omega_{initial})}{4\mu_k r_{pad} \Delta t} \quad (35)$$

Then, using the pressure equation (1), we can claim that the pressure needed in the hydraulic fluid to exert a force N is:

$$P = \frac{N}{A_{brakes}} \quad (36)$$

where A_{brakes} is the surface area of the pistons associated with the brake pads. Furthermore, the force needed to create a pressure from the master cylinder should then be:

$$F = N \frac{A_{master}}{A_{brakes}}$$

where A_{master} is the surface area of the piston associated with the master cylinder of the car. Then, fully expanded out, the force required by the master cylinder to halt the car at some specific initial speed is:

$$F = \frac{\phi_{front} m_{car} (v_{final} - v_{initial}) R + m_{rotor} r_{rotor}^2 (\omega_{final} - \omega_{initial})}{4\mu_k r_{pad} \Delta t} \frac{A_{master}}{A_{brakes}} \quad (37)$$

Finally, we can consider some numerics to solve this qualitatively. Let's assume the *2025 Toyota Camry* situation again, this time modded with *Brembo* brakes. Because the car is Front-Wheel-Drive, its front wheels endure most of the braking force, where $\phi_{front} = 0.8$. The mass of the car and the internal rotor is 1600 kg and 10 kg respectively. The radii of the tires and internal rotor are 0.33 m and 0.15 m respectively, whereas the radius of the brake pad is 0.125 m . Furthermore, $v_{initial}$ and $\omega_{initial}$ have values corresponding to the car moving at 60 mph , so their quantities are 26.8 m/s and 81.2 rad/s respectively. The kinetic friction coefficient associated between the brake pads and internal rotor is 0.4 . Finally, let's assume that A_{master} and A_{brakes} have circular pistons with radii r_{master} and r_{piston} with values 0.012 m and 0.030 m respectively. Area for a circular piston should correspond to the equation πr^2 . Finally, 60 to 0 stoppage for *Brembo's* happens quickly, and Δt can be considered to be 2.2 s . Then, the force exerted by the master cylinder is:

$$F = \frac{0.8(1600\text{ kg})(0 - 26.8\text{ m/s})(0.33\text{ m}) + 10\text{ kg}(0.15\text{ m})^2(0 - 81.2\text{ rad/s})}{4(0.4)(0.125\text{ m})(2.2\text{ s})} \frac{\pi(0.012\text{ m})^2}{\pi(0.030\text{ m})^2}$$

or,

$$F \approx 4120\text{ N}$$

where the negative sign is negated because it is the braking force. If the vacuum booster and brake pedal multiply the force exerted by the foot on the brake pedal by around the 8 times, then the force exerted by the foot in a panic situation is:

$$F_{foot} = 515\text{ N}$$

which is a completely reasonable value, and concludes this section on hydraulic brakes.

2.5 Hydraulic Fluids

Water is a perfectly fine fluid to use for hydraulics since it is incompressible. However, in most if not all industrial applications, hydraulic oil is used, for three main reasons: temperature, lubrication, and corrosion. The subsequent sections will explain these consequences in more depth.

To preface, the hydraulic fluid used most commonly (90% of the time [5]) is mineral oils, see figure 6, will be referred to as a juxtaposition to water.



Figure 6: Mineral oil solutions are used most commonly in hydraulic situations [6].

2.5.1 Temperature

The problem with water pertaining to temperature is that many applications can see heats above the vaporizing temperature, or below the melting temperature. For example, a car's master cylinder will see temperatures in the range of $150\text{ }F^{\circ}$ to $220\text{ }F^{\circ}$, well above the vapor temperature of water. Thus, if water is used in this scenario, the water will spontaneously boil immediately, and there will be no fluid to compress making Pascal's law inapplicable. Gases are compressible because of the wide space between molecules. Since there is no pressure exerted equally and undiminished in all directions of the medium, the brake pistons are never able to see the pressure, and thus, not able to exert an outward force. For general applications, mineral based oils begin to boil at $288\text{ }F^{\circ}$, clearing the temperature of the master cylinder.

On the other end of the extreme, cold-based applications for hydraulics will see negative effects when using water. Construction equipment that utilizes hydraulic systems can freeze during the winter and cause machinery to break down. When water freezes into a lattice grid, its volume increases in certain sections which can apply intense stresses around its enclosure. It's analogous to placing soda in a freezer, and realizing hours later that the walls of the freezer are coated in sugar. Therefore, it is common to use mineral based oil for cold applications, which have specialized polymers that protect microscopic particles from spontaneously freezing. Some hydraulic fluids can withstand $40\text{ }F^{\circ}$, which is around 6° colder than the normal temperature of an arctic winter. Sir William Armstrong, inventor of the hydraulic accumulator, hydraulic crane, and the *Armstrong* gun, realized the pain of water freezing in hydraulics, and attempted

to include additives in the fluid like glycerol to lower the freezing point. However, this did not solve the problem of corrosion, see section 2.5.3, which would break down the surrounding material.

2.5.2 Lubrication

Water can increase friction between moving parts in two ways. One is its polarity. H_2O has a natural electrical polarity that causes it to want to bond to other molecules or atoms. Physics experiments show that this microscopic tendency of attraction causes an overall drag that increases friction between two objects. There is also a phenomena called the meniscus effect. Water has surface tension that forms a tight skin, if this surface tension is present between two close surfaces, the tight skin will create a resistance when those two surfaces attempt to move. Again, these are all microscopic effects, but when multiplied over thousands of times, it can cause a great loss in efficiency.

Mineral based hydraulic fluid contains hydrocarbons, or oils, that do not have polarity and have weaker surface tension. The reason for the latter is that surface tension for hydrocarbons is created by much weaker intermolecular forces, rather than powerful hydrogen bonds for water. Oils also have the effect of moving around dirt in a system, which causes unnecessary abrasiveness regarding moving parts. Because of its non-polar nature, it has the capacity to dissolve other non-polar molecules like grime and dirt surprisingly well, and carry them to a filter to remove them.

2.5.3 Corrosion

The most detrimental effect that water has when used in hydraulics is its oxidation reaction with iron. The way rust forms from iron particles and oxygen particles is from the chemical equation:



Naturally, iron atoms will shave off a steel surface due to the oxygen atoms in H_2O having high polarity and wanting to attach due to high electronegativity. The main problem with rust is that it will flake off over time revealing more layers of fresh iron. This reaction compounds until fractures occur from sections of the steel being smaller and weaker than others, causing huge mechanical failures especially with systems associated moving parts. Hydrocarbons do not garner the same effect because there are no water molecules present to give iron atoms the necessary oxygen atoms to form iron oxide. Furthermore, when a system is supplied with oil, the air molecules outside of the system are effectively sealed off from entering, meaning that no auxiliary oxygen atoms or moisture can enter to produce iron oxide. If one ever tries to mix oil and water, it is clear that oil is extremely hydrophobic, repelling the water due to its non-polar, non-reactive, nature.

3 Components and their Explanations

References

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